

Raspberry Pi: Controlling Eight Servo Motors

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A servo motor is a rotary actuator that allows precise control of angular positions. It consists of a motor coupled to a sensor for position feedback, through a reduction gearbox. These motors are widely used

in applications such as robotics, CNC machinery or automated manufacturing. Servos are controlled by sending an electrical pulse of variable width, or pulse width modulation (PWM) signals, through the control wire.

Servo motors can usually only turn 90 degrees in either direction, totalling 180-degree movement. The motor's neutral position is defined as the position where the servo has the same amount of potential rotation in both the clockwise and counter-clockwise directions.

The PWM signal sent to the motor determines the position of the shaft, and based on the duration of the pulse sent via the control wire, the rotor will turn to the

desired position. The servo motor expects to see a pulse every 20 ms and the length of the pulse will determine how far the motor turns. For example, a 1.5ms pulse will make the motor turn to the 90-degree position (see Fig. 2).

Raspberry Pi has one PWM pin (GPIO 18 or physical pin 12), which can drive only one servo motor at a time with the help of wiring-pi. Presented here is a project with which you can drive up to eight servos together, using Servoblaster, which is a kernel level hardware tweaking program for Raspberry Pi.

Circuit and working

Fig. 3 shows the pin description of the GPIO connector on Raspberry Pi. The GPIO connector has pins with different functions. The device has seven true GPIO pins, an I2C interface, an SPI interface, serial TX/RX pins, and PWM pins that can be used to control external hardware.

Fig. 4 shows the connection scheme

for connecting servo motors to Raspberry Pi's GPIOs. Eight servo motors are interfaced with GPIO 4, 17, 18, 21, 22, 23, 24 and 25 of Raspberry Pi. Refer to Table 1 to identify the physical pin number corresponding to each GPIO to make the connections. An external power supply is used

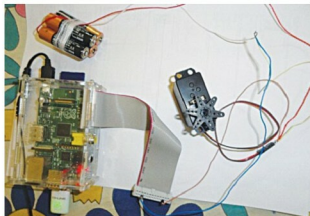


Fig. 1: The author's prototype

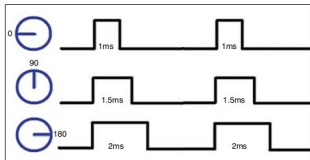


Fig. 2: Pulse width and corresponding servo motor angles

Pin	Function
1	5V
2	Ground
3	GPIO 4
4	GPIO 15
5	GPIO 18
6	GPIO 23
7	GPIO 24
8	GPIO 25
9	GPIO 17
10	GPIO 16
11	GPIO 12
12	GPIO 13
13	GPIO 14
14	GPIO 19
15	GPIO 20
16	GPIO 21
17	GPIO 22
18	GPIO 23
19	GPIO 24
20	GPIO 25
21	GPIO 17
22	GPIO 16
23	GPIO 12
24	GPIO 13
25	GPIO 14
26	GPIO 19
27	GPIO 20
28	GPIO 21
29	GPIO 22
30	GPIO 23
31	GPIO 24
32	GPIO 25

Fig. 3: Pin description of Raspberry Pi

Servo Motors Connection with Raspberry Pi

Servo Number	GPIO	Pin
0	4	7
1	17	11
2	18	12
3	21	13
4	22	15
5	23	16
6	24	18
7	25	22

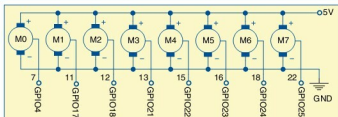


Fig. 4: Connection scheme for servo motors

